

CERTIFICATE



23 June 2026

Ref No: QSPGGN20254878

To Whom It May Concern,

This is to certify that **Mr. Sahil Mehta**, University Roll Number: **2224539**, a student of **I.K. Gujral Punjab Technical University, Kapurthala**, is currently undergoing training and internship as part of the Data Science Course at QSpiders (A Unit of Test Yantra Software Solutions (India) Pvt. Ltd.).

He joined the program on 10th January 2026 and is actively involved in hands-on training and project work in the field of Data Science and Software Development. His training includes key areas such as Python Programming, Data Analysis, Data Visualization, Statistics, Machine Learning, SQL, Database Management, Software Development Fundamentals, Problem Solving, and real-world Data Science projects.

Mr. Sahil Mehta has demonstrated strong learning ability, analytical thinking, attention to detail, and a dedicated approach towards his assignments. He has shown good progress in both technical and practical aspects of Data Science and Software Development. His performance during the training and internship period has been satisfactory.

This certificate is issued upon his request for submission towards his final semester viva examinations as required by his academic program.

We wish him all the best in his future endeavors.

Sincerely,


MR. DEBASIS BARIK
Branch Head

CANDIDATE'S DECLARATION

I, Sahil Mehta, hereby declare that I have undertaken Semester Industrial Training at QSpiders (A Unit of Test Yantra Software Solutions India Pvt. Ltd.) during the period from January 2026 to June 2026 in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Computer Science and Engineering) at I.K. Gujral Punjab Technical University, Kapurthala.

The work presented in this Industrial Training Report entitled “FinGuard AI: Finance and Cyber Security Intelligent Assistant using RAG and LLM” is an authentic record of the work carried out by me during the training period under the guidance of the industry mentors and faculty guide.

I further declare that this report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

Date: _____

Place: _____

Signature of the Student

(Sahil Mehta)

University Roll No.: 2224539

B.Tech (Computer Science & Engineering)

I.K. Gujral Punjab Technical University, Kapurthala

The Semester Industrial Training Viva-Voce Examination of Sahil Mehta has been held on _____ and accepted.

ABSTRACT

Industrial training plays a vital role in bridging the gap between academic learning and industry requirements. During the six-month industrial training program conducted at QSpiders (A Unit of Test Yantra Software Solutions India Pvt. Ltd.), I gained practical exposure to modern software development technologies, programming methodologies, database management systems, web development frameworks, and Artificial Intelligence applications.

As part of the training, I worked on a major project titled **“FinGuard AI: Finance and Cyber Security Intelligent Assistant using RAG and LLM.”** The objective of this project was to develop an intelligent chatbot capable of providing accurate information related to finance and cyber security using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).

The system was designed using Python, FastAPI, Streamlit, ChromaDB, Sentence Transformers, and Groq LLM. The application allows users to ask finance and cyber security related questions in natural language. It retrieves relevant information from a vector database and generates context-aware responses using a Large Language Model. Additionally, the system supports PDF document uploads, enabling users to add custom knowledge sources dynamically.

The project integrates semantic search, vector embeddings, document chunking, conversation memory, and REST APIs to create a scalable and intelligent solution. The developed system improves information retrieval accuracy while reducing the chances of hallucinated responses.

Through this training, I enhanced my technical skills in Python programming, Artificial Intelligence, API development, vector databases, Retrieval-Augmented Generation, and cloud-ready application architecture. The experience provided valuable insights into industry-standard software development practices and strengthened my problem-solving and analytical abilities.

Keywords: Artificial Intelligence, Large Language Model, Retrieval-Augmented Generation, ChromaDB, FastAPI, Streamlit, Semantic Search, Finance, Cyber Security.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **QSpiders (A Unit of Test Yantra Software Solutions India Pvt. Ltd.)** for providing me with the opportunity to undergo Semester Industrial Training and gain practical exposure to modern software development technologies.

I am deeply thankful to my industry mentors, **Mr. Himanshu** and **Ms. Mehak Kumari**, for their valuable guidance, encouragement, and technical support throughout the training period. Their insights and suggestions played an important role in the successful completion of my project.

I would also like to express my heartfelt thanks to my faculty guide, **Dr. Gagandeep Singh**, for his continuous motivation, guidance, and support during the preparation of this training report.

I extend my gratitude to the faculty members of the **Department of Computer Science and Engineering, I.K. Gujral Punjab Technical University, Kapurthala**, for providing the academic knowledge and encouragement required to complete this training successfully.

I am thankful to my friends and fellow trainees for their cooperation and support throughout the training period.

Finally, I express my deepest gratitude to my parents and family members for their constant encouragement, moral support, and belief in my abilities.

I sincerely acknowledge everyone who contributed directly or indirectly to the successful completion of my industrial training and project titled “**FinGuard AI: Finance and Cyber Security Intelligent Assistant using RAG and LLM.**”

Sahil Mehta

University Roll No.: 2224539

B.Tech (Computer Science & Engineering)

I.K. Gujral Punjab Technical University, Kapurthala

ABOUT THE COMPANY / INDUSTRY / INSTITUTE

QSpiders (A Unit of Test Yantra Software Solutions India Pvt. Ltd.)

QSpiders is one of India's leading software training and development organizations, dedicated to providing industry-oriented training in software technologies and preparing students for successful careers in the IT industry. Established in 2007 and headquartered in Bengaluru, Karnataka, QSpiders operates through multiple training centers across India and has trained thousands of students in software development, software testing, full-stack development, and emerging technologies.

QSpiders focuses on bridging the gap between academic learning and industry requirements by providing practical training, real-time projects, and hands-on experience with modern technologies. The institute offers comprehensive training programs in areas such as Java Full Stack Development, Python Full Stack Development, Software Testing, Automation Testing, Data Science, Cloud Computing, DevOps, and Artificial Intelligence.

The training methodology at QSpiders combines theoretical concepts with practical implementation. Students are encouraged to work on real-world projects, participate in coding exercises, attend technical workshops, and gain exposure to industry-standard tools and frameworks. The organization emphasizes problem-solving skills, technical excellence, and professional development.

A major strength of QSpiders is its dedicated placement support system. The institute conducts aptitude training, technical interview preparation, mock interviews, resume-building sessions, and recruitment drives in collaboration with leading IT companies. This helps students improve their employability and secure opportunities in the software industry.

During the industrial training period, I received valuable guidance from experienced trainers and mentors, which helped me strengthen my programming, software development, and Artificial Intelligence skills. The knowledge and practical experience gained through the training played a crucial role in the successful development of my project titled **“FinGuard AI: Finance and Cyber Security Intelligent Assistant using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).”**

Overall, QSpiders provided an excellent learning environment that enhanced both my technical knowledge and professional skills, preparing me for future opportunities in the software and technology industry.

DEFINITIONS, ACRONYMS & ABBREVIATIONS

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
B.Tech	Bachelor of Technology
CISA	Cybersecurity and Infrastructure Security Agency
CVE	Common Vulnerabilities and Exposures
CWE	Common Weakness Enumeration
DB	Database
FD	Fixed Deposit
GDP	Gross Domestic Product
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
ML	Machine Learning
MITRE	MITRE Corporation Knowledge Framework
NVD	National Vulnerability Database
NLP	Natural Language Processing

Dept. of CSE, IKG Punjab Technical University, Kapurthala

Acronym	Full Form
NSE	National Stock Exchange
OTP	One-Time Password
PDF	Portable Document Format
PTU	Punjab Technical University
RAG	Retrieval-Augmented Generation
RBI	Reserve Bank of India
REST	Representational State Transfer
SIP	Systematic Investment Plan
SQL	Structured Query Language
UI	User Interface
UPI	Unified Payments Interface
URL	Uniform Resource Locator
VS Code	Visual Studio Code
WWW	World Wide Web

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